

Heat Transfer Final

Heat Exchangers

$$R_{tot} \triangleq \frac{1}{UA} \Rightarrow \frac{1}{UA} + \frac{1}{h_c A_c} + \frac{R''_{f,c}}{A_c} + R_{w,cond} + \frac{R''_{f,h}}{A_h} + \frac{1}{h_h A_h}$$

(see table 11.1)

$$\dot{m}_h C_{p,h} (T_{h,i} - T_{h,o}) = q = \dot{m}_c C_{p,c} (T_{c,o} - T_{c,i})$$

LMTD

$$q = UA \Delta T_{lm} = UA \frac{\Delta T_2 - \Delta T_1}{\ln(\Delta T_2 / \Delta T_1)}$$

Concentric Parallel Flow

$$\Delta T_1 \triangleq T_{h,i} - T_{c,i} \quad \Delta T_2 \triangleq T_{h,o} - T_{c,o}$$

Concentric Counter Flow

$$\Delta T_1 \triangleq T_{h,o} - T_{c,i} \quad \Delta T_2 \triangleq T_{h,i} - T_{c,o}$$

For the same inlet and outlet temps $\Delta T_{lm\perp} > \Delta T_{lm\parallel}$

NTU

$$\varepsilon \triangleq \frac{q}{q_{max}} \quad q = \varepsilon q_{max} = \varepsilon C_{min} (T_{h,i} - T_{c,i})$$

$$\varepsilon = f(NTU, Cr) \quad Cr \triangleq \frac{C_{min}}{C_{max}} \quad NTU \triangleq \frac{UA}{C_{min}}$$

$$\text{For } Cr = 0 \text{ (all HE)} \Rightarrow \varepsilon = 1 - \exp(-NTU)$$

Radiation

- Reflectivity ρ : Fraction of G that is reflected by the surface
- Absorptivity α : Fraction of G that is absorbed by the surface
- Transmissivity τ : Fraction of G that is transmitted by the surface ($\tau = 0$ in this course)

$$J = E + G_{ref} \quad q''_{rad} = \varepsilon \sigma T_s^4 - \alpha G$$

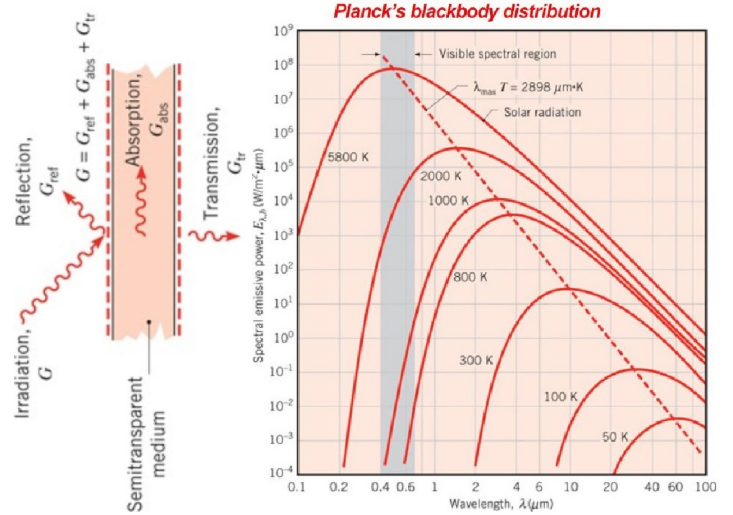
$$d\omega = \sin\theta d\theta d\phi \quad \omega_{hemisphere} = 2\pi \text{ [Sr]}$$

Radiative Intensity

$$I_{\lambda,e}(\lambda, \theta, \phi) = \frac{dq}{dA_1 \cos\theta d\omega d\lambda} \left[\frac{W}{m^2 \cdot Sr \cdot \mu m} \right]$$

Spectral Heat Rate and Flux

$$dq_\lambda = \frac{dq}{d\lambda} = I_{\lambda,e}(\lambda, \theta, \phi) \cos\theta \sin\theta d\theta d\phi \left[\frac{W}{\mu m} \right]$$



$$dq''_\lambda = \frac{dq_\lambda}{dA_1} = I_{\lambda,e}(\lambda, \theta, \phi) \cos\theta d\omega \left[\frac{W}{m^2 \cdot \mu m} \right]$$

Spectral and Total Emissive Power

$$E_\lambda = q''_\lambda = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_{\lambda,e}(\lambda, \theta, \phi) \cos\theta \sin\theta d\theta d\phi \left[\frac{W}{m^2 \cdot \mu m} \right]$$

$$E = \int_{\lambda=0}^{\infty} E_\lambda d\lambda = \varepsilon \sigma T_s^4 \left[\frac{W}{m^2} \right]$$

$$E = \int_{\lambda=0}^{\infty} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_{\lambda,e}(\lambda, \theta, \phi) \cos\theta \sin\theta d\theta d\phi d\lambda$$

$$\text{If the emitter is diffuse} \Rightarrow E_\lambda = \pi I_{\lambda,e}(\lambda)$$

Spectral and Total Irradiation

$$G_\lambda = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_{\lambda,i}(\lambda, \theta, \phi) \cos\theta \sin\theta d\theta d\phi \left[\frac{W}{m^2 \cdot \mu m} \right]$$

$$G = \int_{\lambda=0}^{\infty} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_{\lambda,i}(\lambda, \theta, \phi) \cos\theta \sin\theta d\theta d\phi d\lambda \left[\frac{W}{m^2} \right]$$

$$\text{If the surface is diffuse} \Rightarrow G_\lambda = \pi I_{\lambda,i}(\lambda) \text{ and } G = \pi I_i$$

Spectral and Total Radiosity

$$J_\lambda = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_{\lambda,e+r}(\lambda, \theta, \phi) \cos\theta \sin\theta d\theta d\phi \left[\frac{W}{m^2 \cdot \mu m} \right]$$

$$J = \int_{\lambda=0}^{\infty} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_{\lambda,e+r}(\lambda, \theta, \phi) \cos\theta \sin\theta d\theta d\phi d\lambda \left[\frac{W}{m^2} \right]$$

$$\text{If the reflector and emitter are diffuse} \Rightarrow J_\lambda = \pi I_{\lambda,e+r}(\lambda)$$

Blackbody

- Absorbs **all** incident radiation
- No surface can emit more energy than a blackbody for a prescribed temperature and wavelength
- Blackbody emission = $f(\lambda, T)$ and $\neq f(\theta, \phi)$

$$I_{\lambda,b}(\lambda, T) = \frac{2hc_o^2}{\lambda^5 [\exp(hc_o/\lambda k_B T)^{-1}]} \left[\frac{W}{m^2 \cdot Sr \cdot \mu m} \right]$$

Where:

$h =$	6.626×10^{-34}	$[Js]$	$=$ Planck's Const.
$k_B =$	1.581×10^{-23}	$[J/K]$	$=$ Boltzman's Const.
$c_o =$	2.998×10^8	$[m/s]$	$=$ Speed of light
$T =$	problem subject	$[K]$	$=$ Temperature

$$E_b = \int_{\lambda=0}^{\infty} E_{\lambda,b} d\lambda = \sigma T^4 \quad I_b = \frac{E_b}{\pi} = \frac{\sigma T^4}{\pi}$$

$$\text{Wien's Law} \quad \lambda_{max} T = 2898 \mu m K$$

Band Emission

$$F_{0 \rightarrow \lambda} = \frac{\int_0^{\lambda} E_{\lambda,b} d\lambda}{\int_0^{\infty} E_{\lambda,b} d\lambda} \quad \text{Tabulated, see table 12.2}$$

Radiative Properties

Spectral Directional Emissivity

$$\varepsilon_{\lambda,\theta}(\lambda, \theta, \phi, T) = \frac{I_{\lambda,e}(\lambda, \theta, \phi, T)}{I_{\lambda,b}(\lambda, T)}$$

Spectral Emissivity

$$\varepsilon_{\lambda} = \frac{E_{\lambda}}{E_{\lambda,b}} = \frac{\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_{\lambda,e}(\lambda, \theta, \phi) \cos\theta \sin\theta d\theta d\phi}{\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_{\lambda,b}(\lambda, \theta, \phi) \cos\theta \sin\theta d\theta d\phi}$$

Total Emissivity

$$\varepsilon = \frac{\int_0^{\infty} \varepsilon_{\lambda,b} E_{\lambda,b} d\lambda}{\int_0^{\infty} E_{\lambda,b} d\lambda} = \frac{\int_0^{\infty} \varepsilon_{\lambda,b} E_{\lambda,b} d\lambda}{E_b}$$

Spectral Directional Absorptivity

$$\alpha_{\lambda,\theta}(\lambda, \theta, \phi) = \frac{I_{\lambda,i,abs}(\lambda, \theta, \phi)}{I_{\lambda,i}(\lambda, \theta, \phi)}$$

Spectral Absorptivity

$$\alpha_{\lambda}(\lambda) = \frac{G_{\lambda,abs}(\lambda)}{G_{\lambda}(\lambda)} = \frac{\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} \alpha_{\lambda,\theta} I_{\lambda,i,abs} \cos\theta \sin\theta d\theta d\phi}{\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_{\lambda,i,abs} \cos\theta \sin\theta d\theta d\phi}$$

Diffuse Irradiation

$$\alpha(\lambda, T) = \frac{1}{\pi} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} \alpha(\lambda, \theta) \cos\theta \sin\theta d\theta d\phi$$

If the surface is diffuse $\Rightarrow \alpha_{\lambda} = \alpha_{\lambda,\theta}$

Total Absorptivity

$$\alpha = \frac{G_{abs}}{G} = \frac{\int_0^{\infty} \alpha_{\lambda}(\lambda) G_{\lambda}(\lambda) d\lambda}{\int_0^{\infty} G_{\lambda}(\lambda) d\lambda}$$

Kirchhoff's Law

Always

$$\Rightarrow \varepsilon_{\lambda,\theta} = \alpha_{\lambda,\theta}$$

if (diffuse irradiation OR diffuse surface) $\Rightarrow \varepsilon_{\lambda} = \alpha_{\lambda}$

if (G is blackbody OR surface is GREY) $\Rightarrow \varepsilon = \alpha$

TABLE 11.3 Heat Exchanger Effectiveness Relations [5]

Flow Arrangement	Relation
Parallel flow	$\varepsilon = \frac{1 - \exp[-NTU(1 + C_r)]}{1 + C_r}$
Counterflow	$\varepsilon = \frac{1 - \exp[-NTU(1 - C_r)]}{1 - C_r \exp[-NTU(1 - C_r)]} \quad (C_r < 1)$ $\varepsilon = \frac{NTU}{1 + NTU} \quad (C_r = 1)$
Shell-and-tube	
One shell pass (2, 4, ... tube passes)	$\varepsilon_1 = 2 \left[1 + C_r + (1 + C_r^2)^{1/2} \times \frac{1 + \exp[-(NTU)_1(1 + C_r^2)^{1/2}]}{1 - \exp[-(NTU)_1(1 + C_r^2)^{1/2}]} \right]^{-1}$
n shell passes (2n, 4n, ... tube passes)	$\varepsilon = \left[\left(\frac{1 - \varepsilon_1 C_r}{1 - \varepsilon_1} \right)^n - 1 \right] \left[\left(\frac{1 - \varepsilon_1 C_r}{1 - \varepsilon_1} \right)^n - C_r \right]^{-1}$
Cross-flow (single pass)	
Both fluids unmixed	$\varepsilon = 1 - \exp \left[\left(\frac{1}{C_r} \right) (NTU)^{0.22} \{ \exp[-C_r(NTU)^{0.78}] - 1 \} \right]$
C_{max} (mixed), C_{min} (unmixed)	$\varepsilon = \left(\frac{1}{C_r} \right) (1 - \exp \{ -C_r [1 - \exp(-NTU)] \})$
C_{min} (mixed), C_{max} (unmixed)	$\varepsilon = 1 - \exp \{ -C_r^{-1} [1 - \exp[-C_r(NTU)]] \}$
All exchangers ($C_r = 0$)	$\varepsilon = 1 - \exp(-NTU)$

TABLE 11.4 Heat Exchanger NTU Relations

Flow Arrangement	Relation
Parallel flow	$NTU = -\frac{\ln[1 - \varepsilon(1 + C_r)]}{1 + C_r}$
Counterflow	$NTU = \frac{1}{C_r - 1} \ln \left(\frac{\varepsilon - 1}{\varepsilon C_r - 1} \right) \quad (C_r < 1)$ $NTU = \frac{\varepsilon}{1 - \varepsilon} \quad (C_r = 1)$
Shell-and-tube	
One shell pass (2, 4, ... tube passes)	$(NTU)_1 = -(1 + C_r^2)^{-1/2} \ln \left(\frac{\varepsilon - 1}{\varepsilon + 1} \right)$ $E = \frac{2/\varepsilon_1 - (1 + C_r)}{(1 + C_r^2)^{1/2}}$
n shell passes (2n, 4n, ... tube passes)	Use Equations 11.30b and 11.30c with $\varepsilon_1 = \frac{F - 1}{F - C_r} \quad F = \left(\frac{\varepsilon C_r - 1}{\varepsilon - 1} \right)^{1/n} \quad NTU = n(NTU)_1$
Cross-flow (single pass)	
C_{max} (mixed), C_{min} (unmixed)	$NTU = -\ln \left[1 + \left(\frac{1}{C_r} \right) \ln(1 - \varepsilon C_r) \right]$
C_{min} (mixed), C_{max} (unmixed)	$NTU = -\left(\frac{1}{C_r} \right) \ln[C_r \ln(1 - \varepsilon) + 1]$
All exchangers ($C_r = 0$)	$NTU = -\ln(1 - \varepsilon)$

TABLE 12.2 Blackbody

λT ($\mu m \cdot K$)	$F_{(0 \rightarrow \lambda)}$	λT ($\mu m \cdot K$)	$F_{(0 \rightarrow \lambda)}$
200	0.000000	6,200	0.754140
400	0.000000	6,400	0.769234
600	0.000000	6,600	0.783199
800	0.000016	6,800	0.796129
1,000	0.000321	7,000	0.808109
1,200	0.002134	7,200	0.819217
1,400	0.007790	7,400	0.829527
1,600	0.019718	7,600	0.839102
1,800	0.039341	7,800	0.848005
2,000	0.066728	8,000	0.856288
2,200	0.100888	8,500	0.874608
2,400	0.140256	9,000	0.890029
2,600	0.183120	9,500	0.903085
2,800	0.227897	10,000	0.914199
2,898	0.250108	10,500	0.923710
3,000	0.273232	11,000	0.931890
3,200	0.318102	11,500	0.939959
3,400	0.361735	12,000	0.945098
3,600	0.403607	13,000	0.955139
3,800	0.443382	14,000	0.962898
4,000	0.480877	15,000	0.969981
4,200	0.516014	16,000	0.973814
4,400	0.548796	18,000	0.980860
4,600	0.579280	20,000	0.985602
4,800	0.607559	25,000	0.992215
5,000	0.633747	30,000	0.995340
5,200	0.658970	40,000	0.997967
5,400	0.680360	50,000	0.998953
5,600	0.701046	75,000	0.999713
5,800	0.720158	100,000	0.999905
6,000	0.737818		